



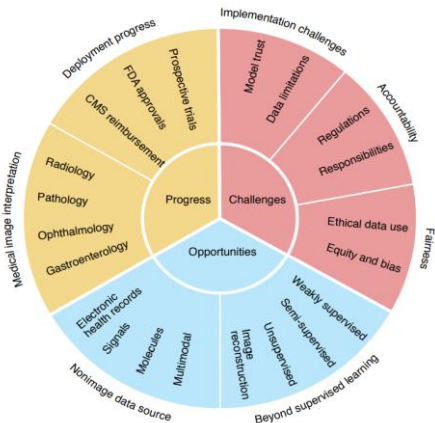
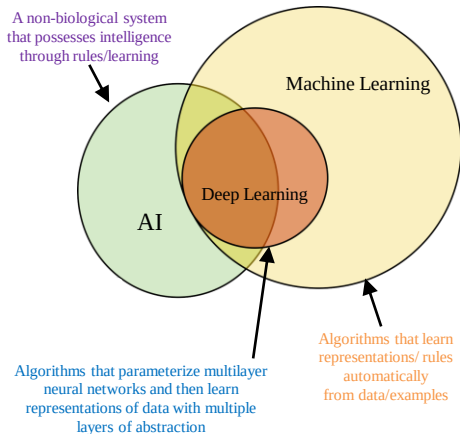
AI for Adaptive Mobile Health

Tanujit Chakraborty



AI in Health and Medicine

“And it (AI) would revolutionize human health, and I think one day maybe we can cure all disease with the help of AI.” - Demis Hassabis, Nobel Laureate of 2024.



Overview of the progress, challenges and opportunities for AI in health*

* Rajpurkar, P., Chen, E., Banerjee, O., & Topol, E. J. (2022). AI in health and medicine. *Nature medicine*, 28(1), 31-38.

Mobile Health (mHealth)

"The smartphone will become the hub of a new era of personalized medicine, transforming the way we monitor and manage our health." - Eric Topol



- The use of mobile and wearable technologies for managing health and wellness (mental health, diabetes, alcohol intake, smoking, physical activity, ...)
- Via personalized and timely messages / push notifications / nudges on mobile phones (Just-in-time adaptive interventions or JITAIs)

Outline

- 1 Microrandomized Trials
- 2 Reinforcement Learning and Thompson Sampling
- 3 Thompson Sampling for Zero-inflated and Overdispersed Count Data
- 4 Discussion

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Micro-randomized Trials (MRTs)

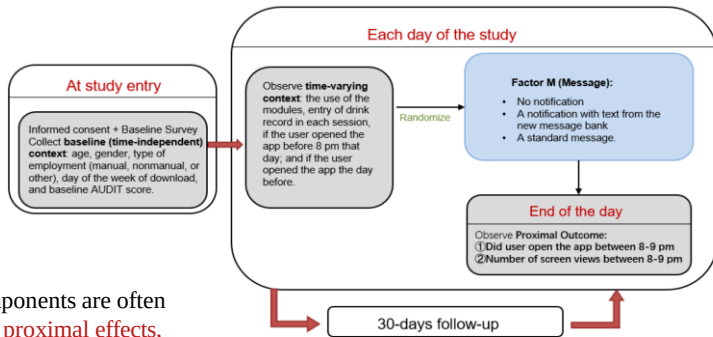
A novel trial design to inform high-quality JITAI construction

- Factorial designs embedded within the Multi-phase Optimization Strategy (MOST) framework are the gold standard when collecting data to build multi-component behavioral interventions.
- MRTs* are cutting-edge trial designs meant to take care of the multi-component, time-varying, sequential nature of the interventions.
 - At each decision stage, randomize between possible interventions.
 - Each person may be randomized at many, many stages (~100s).
 - In case of multi-component interventions, MRTs are sequential, full-factorial designs.

*Liu, X, Nina D, and Bibhas C. "Microrandomized trials: developing just-in-time adaptive interventions for better public health." *American Journal of Public Health* 113.1 (2023): 60-69.

Micro-randomized Trials (MRTs)

A novel trial design to inform high-quality JITAI construction

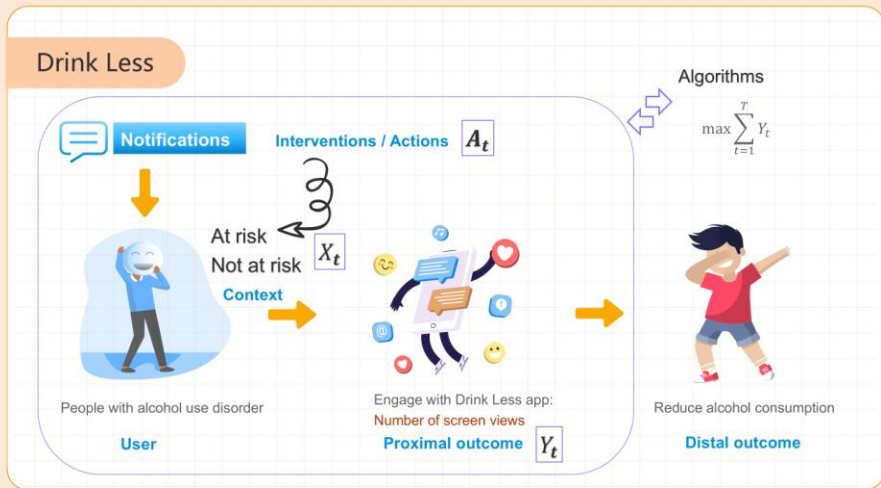


- Intervention components are often intended to have **proximal effects**, e.g., the effect of a motivational message on engagement with an alcohol reduction program or app (**proximal outcome**)
 - Micro-randomization** helps to generate data to accurately assess these **causal** effects.

- Based on scientific theory, the proximal outcomes (e.g., engagement with an alcohol reduction program or app) are expected to eventually impact the **distal outcomes** (e.g., reduced alcohol consumption).

Motivating Example: 'Drink Less' MRT from the UK

People with alcohol use disorder → Engage with Drink Less app → Reduce alcohol consumption



Motivating Example: ‘Drink Less’ MRT from the UK

An mHealth app targeting people who have hazardous alcohol consumption*

- Decision times t : $t = 1, \dots, T$ (=30 days)
- Context X_t : age, gender, time, yesterday's engagement...
- Intervention A_t : Sending versus not sending notifications (binary)
- Proximal Outcome Y_t : **Engagement with the app**, measured by the **number of screen views** in the hour following the notification (8pm - 9pm)
- Distal outcome: Drinking less

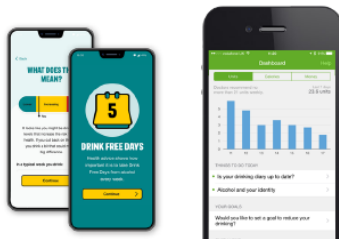


Figure: A behavior change app named *Drink Less*
User-app interaction process: whether the user opens the app, and how much the user uses different modules of the app

* Bell, Lauren, et al. "How notifications affect engagement with a behavior change app: results from a micro-randomized trial." *JMIR mHealth and uHealth* 11.1 (2023): e38342.

MRT Data Structure

It's a bit more than standard longitudinal data!

- T stages (or decision points) on a single individual:

$$X_1, A_1, Y_1, \dots, X_t, A_t, Y_t, \dots, X_T, A_T, Y_T$$

X_t : Context (pre-treatment variables);

A_t : Action (treatment, intervention);

Y_t : Proximal Outcome (Response, Reward).

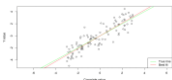
- Many MRTs also involve a variable called **Availability** (denoted I_t) to receive intervention, but we will skip it here for simplicity.

Can we make the MRT design 'adaptive' so that it learns as it progresses?

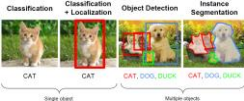
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A Loose Taxonomy of ML

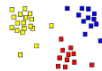


Regression

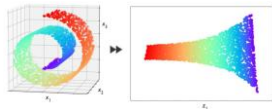


Learning using
labeled data

Learning using
unlabeled data



Clustering

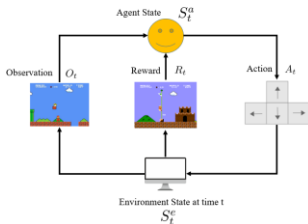


Supervised
Learning

Unsupervised
Learning

Machine
Learning

Reinforcement
Learning



RL doesn't use "labeled" or "unlabeled" data in the traditional sense! In RL, an agent learns via its interactions with an environment

Wish to teach an agent optimal policy for some task

Agent does the following repeatedly

- Senses/observes the environment
- Takes an action based on its current policy
- Receives a reward for that action
- Updates its policy

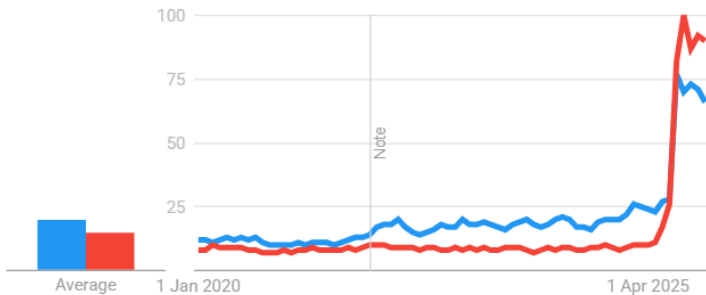
Agent's goal is to maximize its overall reward

Popularity of RL and mHealth

Interest over time

Google Trends

● Reinforcement learning ● Mobile health

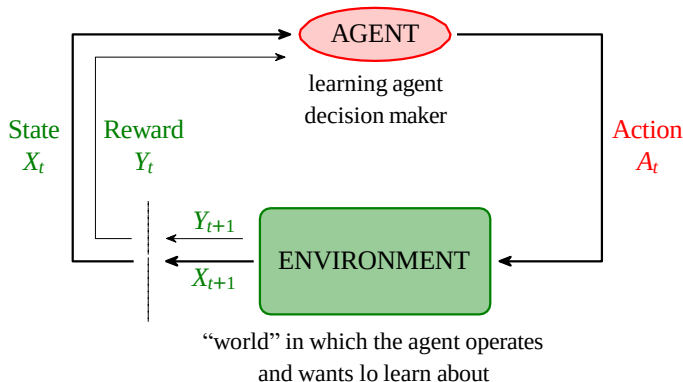


Worldwide. 01/01/2020 - 31/12/2025. Web Search.

Reinforcement Learning (RL)

A Branch of Artificial Intelligence (AI)

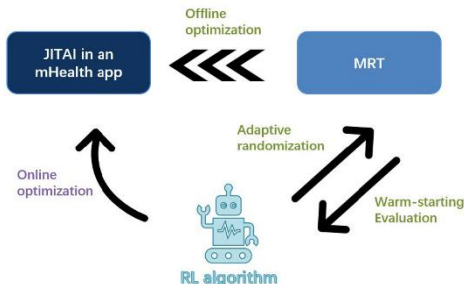
- Learning to take better actions (interventions) in an environment by interacting with it has been studied in the field of **Reinforcement Learning (RL)**, with classical applications in **robotics, automated flights, games**, etc.*



* Sutton, R. S., & Barto, A. G. (1998). *Reinforcement learning: An introduction*. Cambridge: MIT press.

Contextual Bandits

- Short-horizon RL algorithms, like **contextual multi-armed bandit (MAB)** algorithms provide an ideal framework for **customizing JITAs** and/or **adaptive randomization within an MRT** (as well as for **online ads**).



- Exploration-exploitation dilemma**: to select **interventions (actions)** with the highest expected **proximal outcomes (rewards)** based on current estimates, or to explore other interventions that may improve the estimates of expected proximal outcomes?

(Linear) Thompson Sampling

One Popular Contextual Bandit Algorithm

- **Thompson Sampling (TS)** (*Thompson, 1933, Biometrika*) is a multi-step **Bayesian** algorithm that:
 1. specifies a **reward (outcome)** model;
 2. specifies **prior distributions** for the model
 3. parameters; at a given step:
 - computes a posterior distribution of the parameters based on observed data, and then selects the action that **maximizes the expected posterior reward at the given context**;
 - generates corresponding reward data;
 4. repeats the above steps.
- The most common version is **Linear Thompson Sampling (LinTS)** (*Agrawal and Goyal, 2013, ICML*) suitable for **continuous, symmetric outcome distributions**.

(Linear) Thompson Sampling

One Popular Contextual Bandit Algorithm

- Thompson Sampling (TS) is a popular algorithm in mobile health. Let's consider a simple regression model of this form, and ϕ is a feature vector, and β is the model parameter.

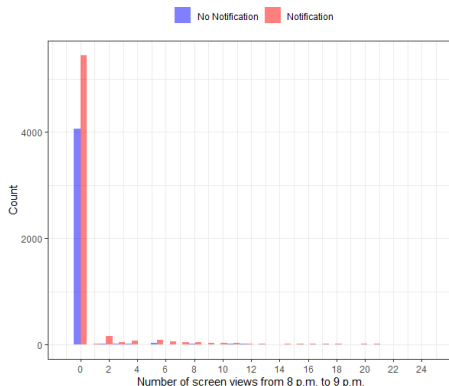
$$E[Y_t | X_t, A_t] = \phi(A_t, X_t)^T \beta$$

- Assume a prior $P_0(\beta)$
 - Using historical data $D_{t-1} = \{X_i, A_i, Y_i; i = 1, \dots, t-1\}$ up to time t to update the posterior $Q_t = P(\beta | D_{t-1})$
 - Sample $\tilde{\beta} \sim Q_t$
 - Observes the current context X_t
 - Deliver the action $A_t = \arg \max_{A_t \in \mathcal{A}} \phi(A_t, X_t)^T \tilde{\beta}$
-  Distribution of Y_t
- Instead of sticking to the **Maximum Likelihood Estimate (or MLE)** for β , it draws a sample from this updated posterior, introduces a degree of randomness, which is pivotal for exploration
 - The goal is to select an action that optimizes the expected outcome. In real world applications, the outcome Y_t might be **binary data or count data**.

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Zero-inflated Count Data from the Drink Less MRT



Methodological Research Problem: We noticed a lack of algorithms for potentially **zero-inflated** and/or **overdispersed** count outcomes in the literature, and chose to address this gap.

Count Data Models

NB with unknown r , ZIP and ZINB do not belong to the Generalized Linear Model (GLM) family!

- **Poisson** regression:

$$E[Y_t|X_t, A_t] = \exp(\phi(A_t, X_t)^T \beta^*), Y_t \sim \text{Poisson}(E[Y_t|X_t, A_t])$$

- **Negative Binomial (NB)** regression:

$$E[Y_t|X_t, A_t] = \exp(\phi(A_t, X_t)^T \beta^*), Y_t \sim \text{NB}(E[Y_t|X_t, A_t], r), \text{ where } r \text{ refers to the dispersion parameter}$$

- **Zero-Inflated Poisson (ZIP)** regression:

$$E[Y_t|X_t, A_t] = (1 - \text{sigmoid}(\phi(A_t, X_t)^T \gamma^*)) \exp(\phi(A_t, X_t)^T \beta^*), Y_t = Z_t C_t, \\ \text{where } Z_t \sim \text{Bernoulli}(1 - \text{sigmoid}(\phi(A_t, X_t)^T \gamma^*)) \text{ and} \\ C_t = \text{Poisson}(\exp(\phi(A_t, X_t)^T \beta^*))$$

- **Zero-Inflated Negative Binomial (ZINB)** regression:

$$E[Y_t|X_t, A_t] = (1 - \text{sigmoid}(\phi(A_t, X_t)^T \gamma^*)) \exp(\phi(A_t, X_t)^T \beta^*), Y_t = Z_t C_t, \\ \text{where } Z_t \sim \text{Bernoulli}(1 - \text{sigmoid}(\phi(A_t, X_t)^T \gamma^*)) \text{ and} \\ C_t = \text{NB}(\exp(\phi(A_t, X_t)^T \beta^*), r)$$

TS for Count Data: Some Challenges and Remedy

- **Challenges:**

1. The posteriors derived from the likelihood functions of count data models and the normal prior are **intractable** due to **lack of conjugacy**.
2. Approximate sampling from these posteriors via Markov chain Monte Carlo (MCMC) may be rather **time-consuming**.

- **Remedy:** Instead of sampling from the intractable posteriors, **we sample $\tilde{\beta}_t$ from a normal distribution $\mathcal{N}(\hat{\beta}_t, \alpha^2 I_t^{-1})$** , where $\hat{\beta}_t$ is the maximum likelihood estimator (MLE), I_t is the Fisher information, and α is a tuning parameter (**Laplace Approximation**).
- **Justification:** Any (posterior) distribution that permits TS to **explore** “far enough” from the MLE $\hat{\beta}_t$ (**anti-concentration**) but “not too much” (**concentration**) can theoretically guarantee “good performance” (**small regret**) (*Abeille and Lazaric, 2017, AISTATS*).

Approximate Sampling for TS-Count

For data **Without Zero-inflation**

Algorithm 1: TS-Count (Poisson / NB)

Input: Tuning parameter α_β , initial random exploration time τ

Initialization: Randomly choose $A_t \in \mathcal{A}$ for $t \leq \tau$ and collect the observations D_τ

for $t = \tau + 1, \dots, T$ **do**

 Compute the MLE $\hat{\beta}_t$

 Get $\mathcal{J}_t(\hat{\beta}_t)$ as a byproduct of step 3

 Sample $\tilde{\beta}_t$ from $\mathcal{N}(\hat{\beta}_t, \alpha_\beta^2 \mathcal{J}_t^{-1}(\hat{\beta}_t))$

 Observe context X_t and compute $A_t = \arg \max_{a \in \mathcal{A}} \exp\{\phi(a, X_t)^\top \tilde{\beta}_t\}$

 Offer intervention A_t and observe the associated outcome Y_t

Approximate Sampling for TS-Count (Cont'd)

For data **With Zero-inflation**

Algorithm 2: TS-Count (ZIP / ZINB)

Input: Tuning parameters α_β and α_γ , initial random exploration time τ

Initialization: Randomly choose $A_t \in \mathcal{A}$ for $t \leq \tau$ and collect the observations D_τ

for $t = \tau + 1, \dots, T$ **do**

 Compute the MLEs $\hat{\gamma}_t$ and $\hat{\beta}_t$

 Get $\mathcal{I}_t(\hat{\gamma}_t)$ and $\mathcal{I}_t(\hat{\beta}_t)$ as a byproduct of steps 3 and 4

 Sample $\tilde{\gamma}_t$ from $\mathcal{N}(\hat{\gamma}_t, \alpha_\beta^2 \mathcal{I}_t^{-1}(\hat{\gamma}_t))$

 Sample $\tilde{\beta}_t$ from $\mathcal{N}(\hat{\beta}_t, \alpha_\beta^2 \mathcal{I}_t^{-1}(\hat{\beta}_t))$

 Observe context X_t and compute

$$A_t = \arg \max_{a \in \mathcal{A}} \exp[1 - \text{sigmoid}(\phi(a, X_t)^\top \tilde{\gamma}_t)] \exp\{\phi(a, X_t)^\top \tilde{\beta}_t\}$$

 Offer intervention A_t and observe the associated outcome Y_t

Performance Metrics of the Learning Algorithms

Regret: “Oh, I wish I could do better, like the Oracle...”

Expected difference in outcomes between the Oracle and a candidate algorithm

- (Cumulative) Frequentist Regret:

$$\mathcal{R}(T, \eta^*) = \sum_{t=1}^T \mathbb{E} \left[\max_{a \in \mathcal{A}} h(\phi(a, X_t)^\top; \eta^*) - h(\phi(A_t, X_t)^\top; \eta^*) \mid \eta^* \right]$$

- (Cumulative) Bayesian Regret:

$$\mathcal{R}_{Bayes}(T) = \sum_{t=1}^T \mathbb{E} \left[\mathbb{E} \left[\max_{a \in \mathcal{A}} h(\phi(a, X_t)^\top; \eta^*) - h(\phi(A_t, X_t)^\top; \eta^*) \mid \eta^* \right] \right]$$

where η^* is the true parameter, i.e., $\eta^* = \beta^*$ for Poisson and NB, and $\eta^* = (\beta^*, \gamma^*)$ for ZIP and ZINB; and $h(\cdot)$ is the inverse-link function, e.g., $h(\cdot) = \exp(\cdot)$.

Theoretical Performance Guarantees

For bandit algorithms, the popular culture is to bound the regrets!

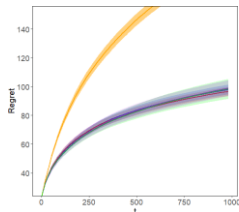
- Regret bounds for TS with the GLM family of distributions (including the **Poisson** model) already exists.
- However: **NB with unknown dispersion parameter (r)**, ZIP and ZINB do not belong to the GLM family!
- **Our contribution:** Under standard conditions (e.g., bounded rewards and feature space), we have proved:
 - $O(d\sqrt{T})$ **Bayesian regret bound** for TS-Count, which coincides with the regret bound for generalized linear TS (*Russo and Van Roy, 2014*)
 - $O(d\sqrt{T\log K})$ **frequentist regret bound** for TS-Count, which coincides with the regret bound for generalized linear TS (*Kveton et al., 2020*)
 - Theoretical derivations provided us with an upper bound for the regret of these algorithms, illustrating that their regret escalates at a sublinear rate over time, T .

where d : dimension of the feature vector, T : time horizon, K : number of intervention options.

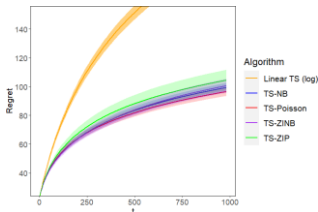
Simulation Results (Frequentist Regrets)

True Data Generative Model: Overdispersed Poisson

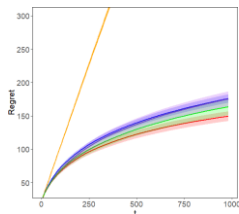
$T = 1000; K = 20; d = 4$; number of replications = 200



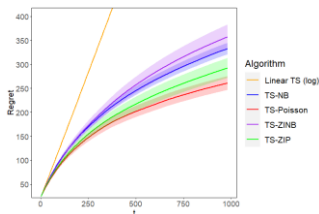
(a) No Overdispersion



(b) Low Overdispersion



(c) Moderate Overdispersion

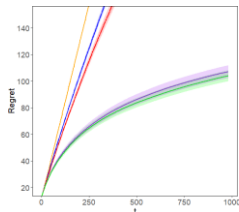


(d) High Overdispersion

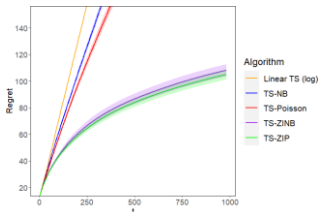
Simulation Results (Frequentist Regrets)

True Data Generative Model: Zero-inflated (ZI) Overdispersed Poisson

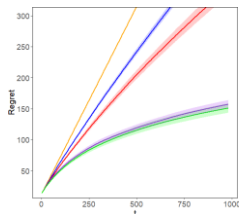
$T = 1000; K = 20; d = 4$; number of replications = 200



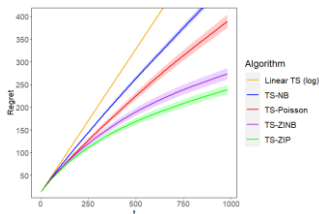
(e) ZI + No Overdispersion



(f) ZI + Low Overdispersion



(g) ZI + Moderate Overdispersion



(h) ZI + High Overdispersion

Comparison with MCMC-based TS-Count

True Data Generative Model: Zero-inflated (ZI) Overdispersed (Moderate) Poisson

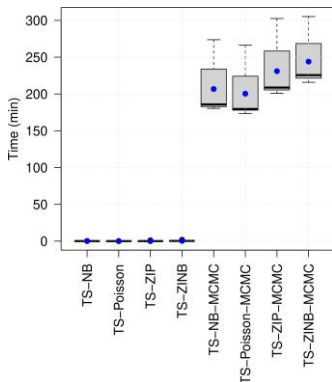


Figure: Computation Time

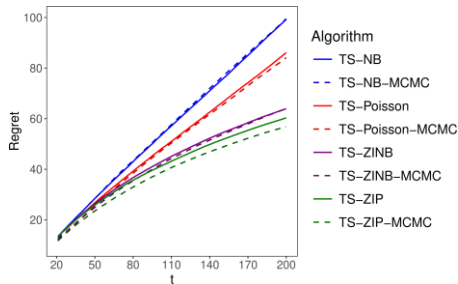


Figure: Regret

Proposed algorithms vs. their MCMC counterparts. Results based on 200 replications.

Case study: Drink Less

Data Generating Process

Environment

- Build a simulation environment based on Drink Less
- Actions: providing notification vs. not providing notification
- Outcome generated from a ZIP model
- True parameters are determined by:
 - Fit a generalized linear mixed model with ZIP distribution
 - Get user-specific regression coefficients

Multi-task problem

- Execute the proposed algorithms individually for each user

After-study analysis

- Clipping the randomization probability

$$\tilde{p}_{a,n,t} = \min(0.95, \max(p_{a,n,t}, 0.05))$$

where $p_{a,n,t}$ can be calculated from the algorithm

Application to Drink Less MRT

A simulation study based on the real data

- Re-design the Drink Less trial, considering **adaptive** versions in addition to the original **non-adaptive MRT**.
- In the absence of zero inflation (left panel), Linear TS does well initially (**too little data for more complex models!**) but TS-count starts to dominate later.

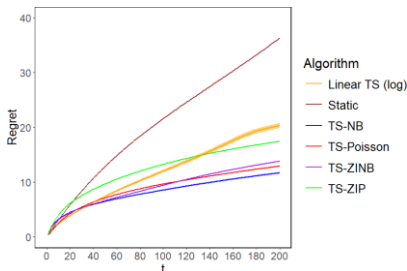


Figure: Overdispersed Poisson Generative Model

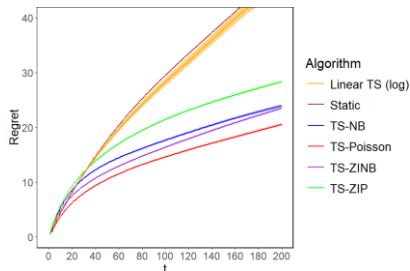


Figure: Zero-inflated Overdispersed Poisson Generative Model

Regrets averaged over 349 users and 100 simulated trials

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Summary of Findings

- MRTs are cutting-edge trial designs in mHealth that can be made **adaptive by TS or other RL** algorithms, thereby improving the trial participants' outcomes.
- Validated methodologies (and software) are available for **zero-inflated count outcomes** from (adaptive) MRTs, in addition to **continuous and binary outcomes**.
- Complex models like **NB and ZINB** (having mechanisms to **accommodate overdispersion**) may not outperform simpler models like **Poisson and ZIP** in **online learning settings like mHealth**, likely due to limited data early on.

Future Works

- Causal inference on intervention components from adaptively randomized data based on TS-Count (*Deliu et al., 2021; Zhang et al., 2023*)
- Incorporating mechanisms for information borrowing between users to increase efficiency (*Tomkins et al., 2021*)
- Exploring alternative approximation approaches for posterior sampling (*Xu et al., 2022*)

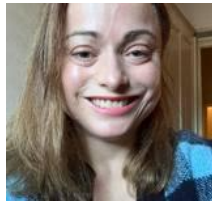
Team and Publication



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R Package

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AI in Health Machine Learning & Forecasting

mHealth

Data

Micro-Randomized Trials
Dynamic treatment regime
Causal Inference



Epidemic TS

centric

Disease Forecasting
Risk assessment
Spatiotemporal Disease
modeling



Interpretable ML

view

Survival Analysis on EHR
Healthcare Data Privacy
Explainable and
Generative AI



Thank You!



essentially,
all models are wrong,
but some are useful

George E. P. Box



SCAN ME